2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.241 Introduction to Computer Graphics

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

Question 1

1.a What is texture mapping and what is the purpose of using it?

[2 marks]

1.b Describe the key ideas behind the pixel-based texture synthesis method and the patch-based texture synthesis method. What is the main difference between them?

[3 marks]

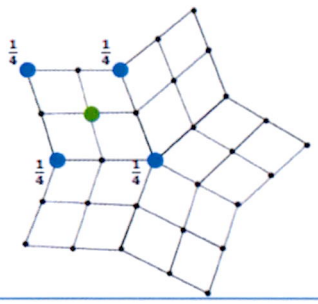
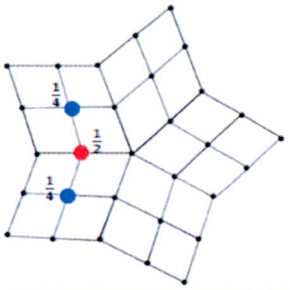
1.c Points A=(1,2,3) and B=(4,5,6) are points that form a parametric line. Is point C=(3,4,5) on this parametric line? If not, explain why not? If it is, what is the value of parameter *t* that gives point C in the parametric line AB?

[4 marks]

1.d Describe the three main rules that are used in Reynold's flocking model. Discuss how the concept of flocking rules may be used to create the motions for a flock of birds with one leader and a large number of followers.

[5 marks]

1.e The rules below are for applying Linear Subdivision. We start with a cube and apply Linear Subdivision to the whole cube. After one iteration, how many quadrilateral faces are there in the result? In the process of applying this first iteration, how many times do we use each of the two rules below?

Geometry refinement rule (for new vertex)	**Geometry refinement rule (for new vertex on existing edge)**
	

[6 marks]

[Total 20 marks]

Question 2

2.a What is the Phong shading model and how does it work? What is the Flat shading model and how does it work?

[4 marks]

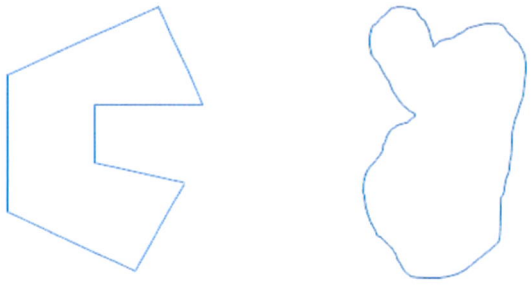
2.b Give an example of a 3D shape that would be appropriate for representing with a parametric surface (and explain why). Give an example of a different 3D shape that would be appropriate for representing with an implicit surface (and explain why).

[4 marks]

2.c The Cyclic Coordinate Descent method for inverse kinematics may fail to work in some cases. Think of a scenario in which it would fail. Explain what the method tries to do in your scenario and explain why it fails.

[3 marks]

2.d For the two 2D shapes below, which one is better represented with a BSP Tree (and which one would *not* be well presented with a BSP Tree)? Explain your answers.



[3 marks]

2.e We have studied many methods of 3D object representation. Assume you wish to create a virtual 3D model of a real-world table. Choose three methods of 3D object representation that can be appropriate for modelling the table. For each method, explain why it is an appropriate representation.

[6 marks]

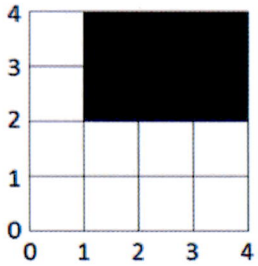
[Total 20 marks]

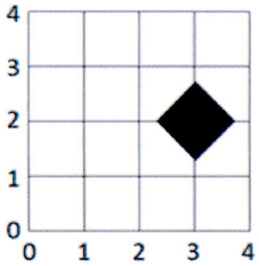
Question 3

3a. What is the key difference in using "bump mapping" to render an object compared to rendering in the usual case? What is one application where this would be useful?

[2 marks]

3b. Write the sequence of OpenGL 2D transformations that maps the given shape on the left (see images below) to the shape on the right. You can use a combination of these functions: glTranslatef(tx, ty, 0.0), glRotatef(angle, 0.0, 0.0, 1.0), and/or glScalef(sx, sy, 1.0).





[3 marks]

3c. Describe how optical motion capture works, and give one advantage and one disadvantage of this technique.

[4 marks]

3d. Which of the 3D object representations that we studied is appropriate for representing each of the following? Explain your answer in each case.

(i) Human face
(ii) Bookshelf
(iii) Helicopter
(iv) University campus

[4 marks]

3e. You are given a ray with the equation P = (1,2,3) + t * (1,1,0), and a sphere with the equation $|P - (8,2,4)|^2 = 5^2$. Perform a ray-sphere intersection. Does the ray intersect with the sphere? If so, what is the value of the closest t? Show your workings.

[7 marks]

[Total 20 marks]

Question 4

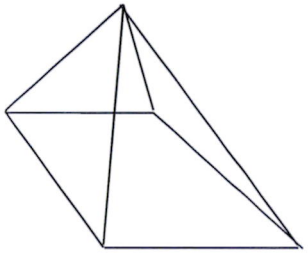
4a. What are the key characteristics of a directional light, a point light, and a spot light?

[3 marks]

4b. Give a *real-life* example of a joint that has one degree-of-freedom. Give a *real-life* example that has two degrees-of-freedom and a *real-life* example that has three degrees-of-freedom. Explain your answer in each case.

[3 marks]

4c. The figure below shows a three-dimensional rectangular-based pyramid (i.e. with rectangle at bottom). Write a vertex table and a face table for this shape. The exact values of the vertices are not important as long as they form the overall shape.



[6 marks]

4d. You are given a ray with the equation P = (1,3,5) + *t* * (2,1,2), and a triangle with vertices at T_1=(3,3,3), T_2=(3,5,7), and T_3=(3,3,9). Perform a ray-triangle intersection by first doing a ray-plane intersection. What is the intersection point for the ray-plane intersection? Is this intersection point inside the given triangle? Show your workings.

[8 marks]

[Total 20 marks]

---End of Paper---